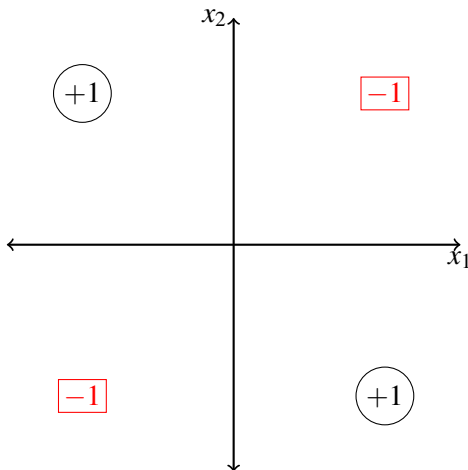


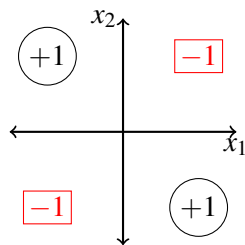
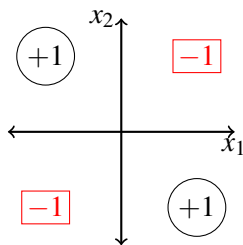
Problema XOR con 3 neuronas

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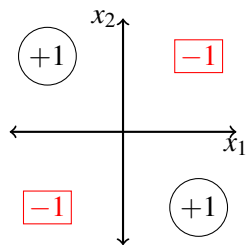
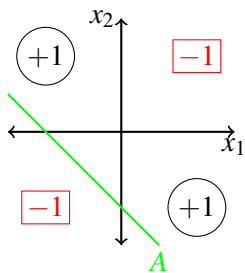
¿Cómo podríamos resolver el problema XOR?



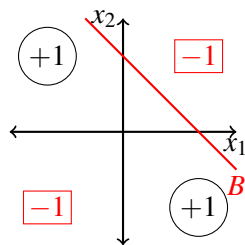
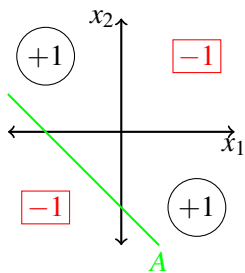
Primera capa: perceptrones A y B



Primera capa: perceptrones A y B



Primera capa: perceptrones A y B



Segunda capa: perceptron C

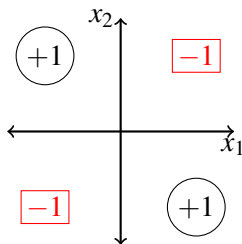
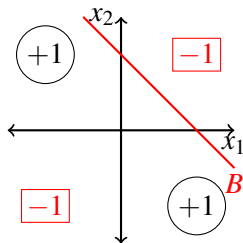
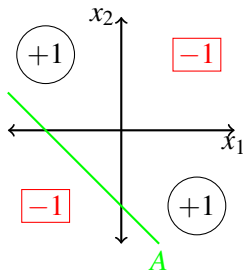


Tabla de verdad para el perceptron C

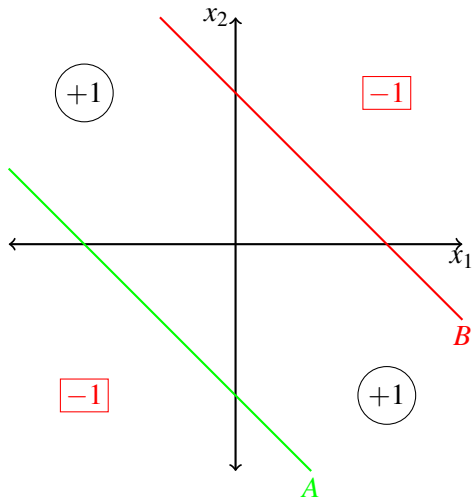


Tabla de verdad para el perceptron C

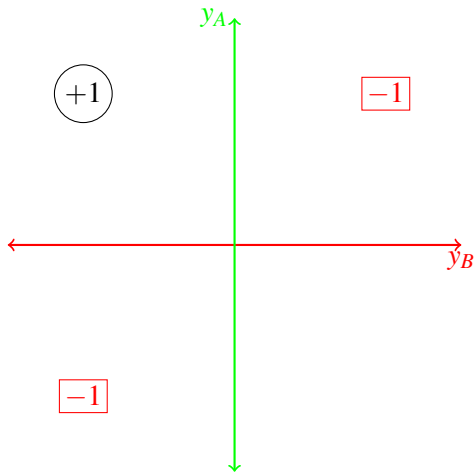


Tabla de verdad para el perceptron C

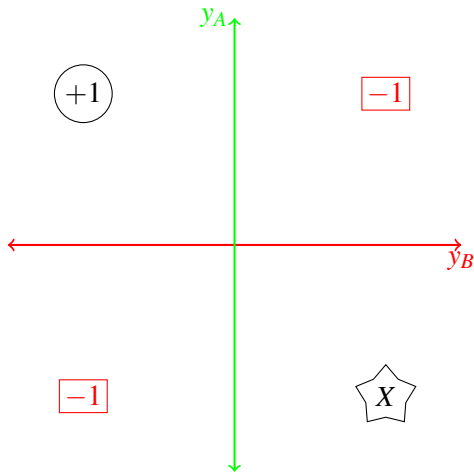
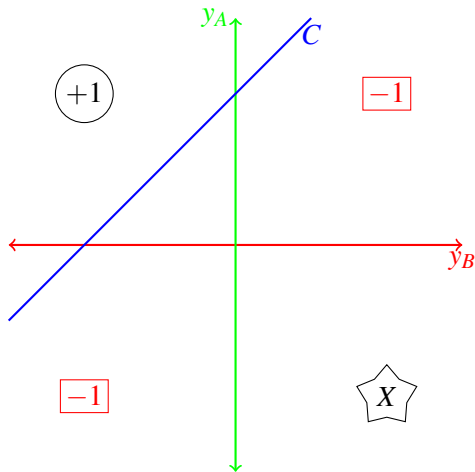


Tabla de verdad para el perceptron C



Nuestra primera red neuronal

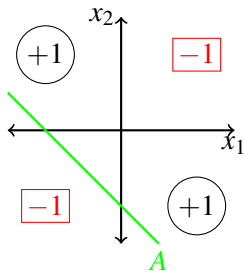
Un perceptrón multicapa con 3 neuronas

Diego Milone

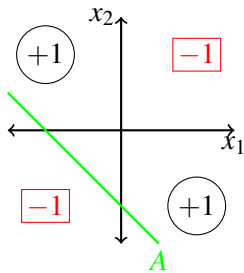
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Combinación de perceptrones simples: A

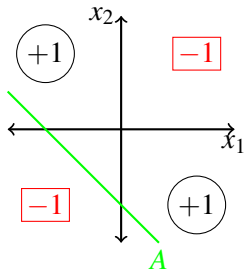


Combinación de perceptrones simples: A



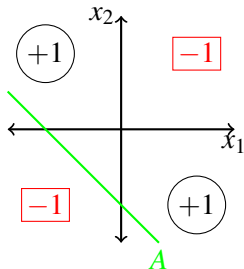
Perceptrón A: $x_2 = -1 - x_1$

Combinación de perceptrones simples: A



Perceptrón A: $x_2 = -1 - x_1 = \frac{w_{A0}}{w_{A2}} - \frac{w_{A1}}{w_{A2}}x_1$

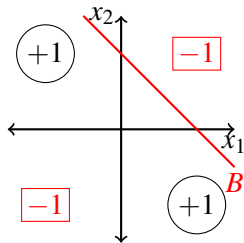
Combinación de perceptrones simples: A



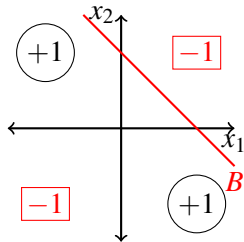
Perceptrón A: $x_2 = -1 - x_1 = \frac{w_{A0}}{w_{A2}} - \frac{w_{A1}}{w_{A2}}x_1$

$$\rightarrow \left\{ \begin{array}{l} w_{A0} = -1 \\ w_{A1} = +1 \\ w_{A2} = +1 \end{array} \right\} \rightarrow y_A = \text{sgn}(x_2 + x_1 + 1)$$

Combinación de perceptrones simples: B

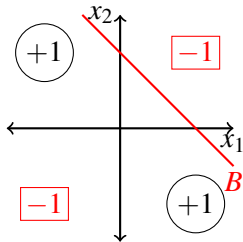


Combinación de perceptrones simples: B



Perceptrón B: $x_2 = +1 - x_1$

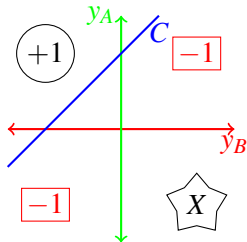
Combinación de perceptrones simples: B



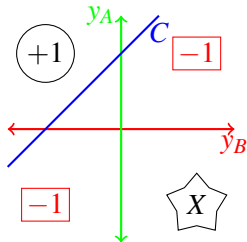
Perceptrón B: $x_2 = +1 - x_1$

$$\rightarrow \left\{ \begin{array}{l} w_{B0} = +1 \\ w_{B1} = +1 \\ w_{B2} = +1 \end{array} \right\} \rightarrow y_B = \text{sgn}(x_2 + x_1 - 1)$$

Combinación de perceptrones simples: C

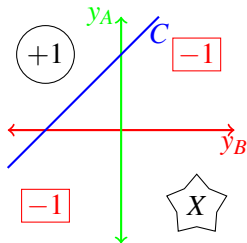


Combinación de perceptrones simples: C



Perceptrón C: $y_A = +1 + y_B$

Combinación de perceptrones simples: C



Perceptrón C: $y_A = +1 + y_B$

$$\rightarrow \left\{ \begin{array}{l} w_{C0} = +1 \\ w_{C1} = -1 \\ w_{C2} = +1 \end{array} \right\} \rightarrow y_C = \text{sgn}(y_A - y_B - 1)$$

¿Cómo es la arquitectura de esta red neuronal?

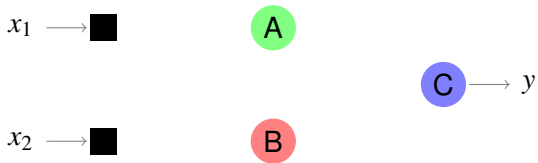
$$\left\{ \begin{array}{l} w_{C0} = +1 \\ w_{C1} = -1 \\ w_{C2} = +1 \end{array} \right\}$$

$$\left\{ \begin{array}{l} w_{A0} = -1 \\ w_{A1} = +1 \\ w_{A2} = +1 \end{array} \right\} \left\{ \begin{array}{l} w_{B0} = +1 \\ w_{B1} = +1 \\ w_{B2} = +1 \end{array} \right\}$$

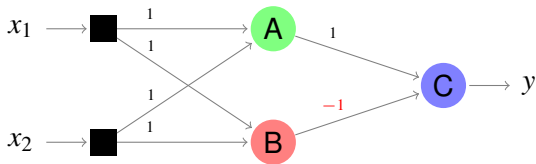
$$\left. \begin{array}{l} y_A = \text{sgn}(x_2 + x_1 + 1) \\ y_B = \text{sgn}(x_2 + x_1 - 1) \end{array} \right\} \rightarrow y_C = \text{sgn}(y_A - y_B - 1)$$

Y... ¿resolverá el XOR?

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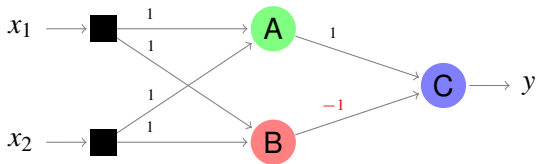


Y... ¿resolverá el XOR?

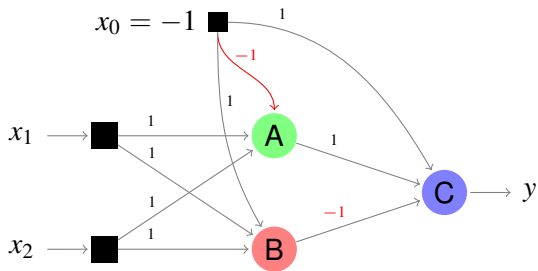


Y... ¿resolverá el XOR?

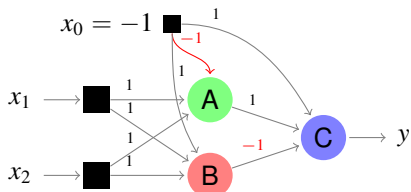
$$x_0 = -1 \blacksquare$$



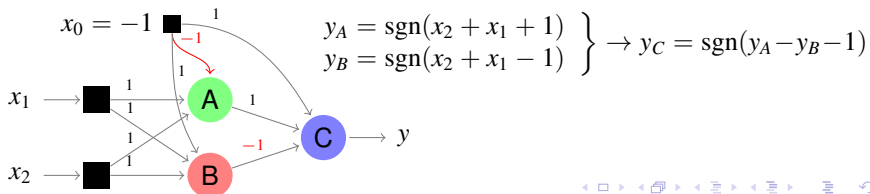
Y... ¿resolverá el XOR?



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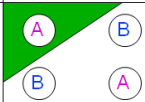
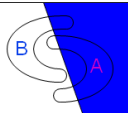
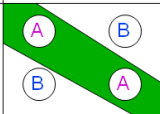
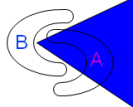
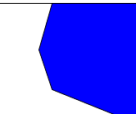
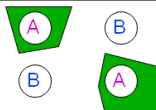
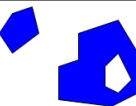


Perceptrón multicapa: regiones de decisión y arquitectura

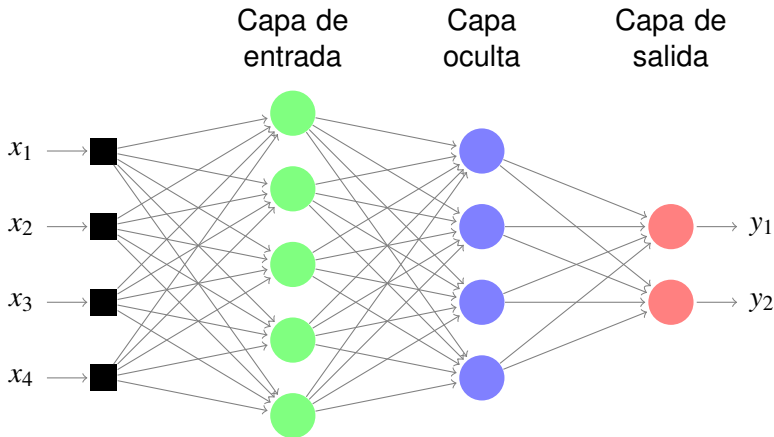
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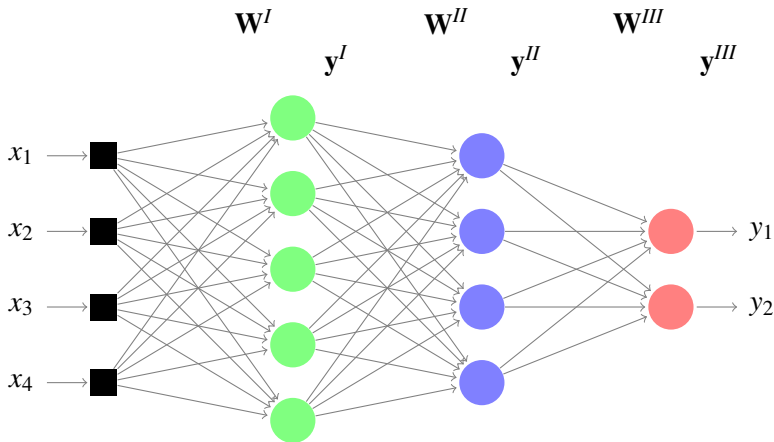
Regiones de decisión

<i>Estructura</i>	<i>Tipos de regiones de decisión</i>	<i>Problema XOR</i>	<i>Separación en clases</i>	<i>Formas regiones más generales</i>
<i>Una capa</i> 	<i>hemiplano limitado por hiperplano</i>			
<i>Dos capas</i> 	<i>Regiones convexas abiertas o cerradas</i>			
<i>Tres capas</i> 	<i>Arbitrarias (Complejidad limitada por N°. de Nodos)</i>			

Arquitectura del perceptrón multicapa



Arquitectura del perceptrón multicapa



Cálculo de las salidas en cada capa

- Capa I:

$$v_j^I = \langle \mathbf{w}_j^I, \mathbf{x} \rangle = \sum_{i=0}^N w_{ji}^I x_i \quad (\text{completo } \mathbf{v}^I = \mathbf{W}^I \mathbf{x})$$

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$$y_j^I = \phi(v_j^I) = \frac{2}{1 + e^{-bv_j^I}} - 1 \quad (\text{simétrica } \pm 1)$$

- Capa II:

$$v_j^{II} = \langle \mathbf{w}_j^{II}, \mathbf{y}^I \rangle \rightarrow y_j^{II} = \phi(v_j^{II})$$

- Capa III:

$$v_j^{III} = \langle \mathbf{w}_j^{III}, \mathbf{y}^{II} \rangle \rightarrow y_j^{III} = \phi(v_j^{III}) = y_j$$

Propagación hacia atrás: caso general y capa de salida

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Criterio de error

Suma del error cuadrático instantáneo

$$\xi(n) = \frac{1}{2} \sum_{j=1}^M e_j^2(n)$$

Aplicación del gradiente (caso general)

$$\Delta w_{ji}(n) = -\mu \frac{\partial \xi(n)}{\partial w_{ji}(n)}$$

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$$\Delta w_{ji}(n) = -\mu \frac{\partial \xi(n)}{\partial w_{ji}(n)}$$

$$\frac{\partial \xi(n)}{\partial w_{ji}(n)} = \frac{\partial \xi(n)}{\partial e_j(n)} \frac{\partial e_j(n)}{\partial y_j(n)} \frac{\partial y_j(n)}{\partial v_j(n)} \frac{\partial v_j(n)}{\partial w_{ji}(n)}$$

Aplicación del gradiente (caso general)

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$$\frac{\partial \xi(n)}{\partial w_{ji}(n)} = \frac{\partial \xi(n)}{\partial e_j(n)} \frac{\partial e_j(n)}{\partial y_j(n)} \frac{\partial y_j(n)}{\partial v_j(n)} \boxed{\frac{\partial v_j(n)}{\partial w_{ji}(n)}}$$

$$\frac{\partial v_j(n)}{\partial w_{ji}(n)} = \frac{\partial \sum_{i=0}^N w_{ji}(n) y_i(n)}{\partial w_{ji}(n)} = y_i(n)$$

Aplicación del gradiente (caso general)

$$\Delta w_{ji}(n) = -\mu \frac{\partial \xi(n)}{\partial w_{ji}(n)}$$

$$\frac{\partial \xi(n)}{\partial w_{ji}(n)} = \left[\frac{\partial \xi(n)}{\partial e_j(n)} \frac{\partial e_j(n)}{\partial y_j(n)} \frac{\partial y_j(n)}{\partial v_j(n)} \right] y_i(n)$$

Gradiente de error local instantáneo: $\delta_j = \frac{\partial \xi(n)}{\partial y_j(n)} \frac{\partial y_j(n)}{\partial v_j(n)}$

Aplicación del gradiente (caso general)

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Gradiente de error local instantáneo: $\delta_j = \frac{\partial \xi(n)}{\partial y_j(n)} \boxed{\frac{\partial y_j(n)}{\partial v_j(n)}}$

Derivada de la función de activación simétrica (1/2)

$$\begin{aligned}
 \frac{\partial y_j(n)}{\partial v_j(n)} &= \frac{\partial \left\{ \frac{2}{1+e^{-v_j(n)}} - 1 \right\}}{\partial v_j(n)} \\
 &= 2 \frac{e^{-v_j(n)}}{(1+e^{-v_j(n)})^2} \\
 &= 2 \frac{1}{1+e^{-v_j(n)}} \frac{e^{-v_j(n)}}{1+e^{-v_j(n)}} \\
 &= 2 \frac{1}{1+e^{-v_j(n)}} \frac{\overbrace{-1+1}^0 + e^{-v_j(n)}}{1+e^{-v_j(n)}} \\
 &= 2 \frac{1}{1+e^{-v_j(n)}} \left(\frac{-1}{1+e^{-v_j(n)}} + \frac{1+e^{-v_j(n)}}{1+e^{-v_j(n)}} \right)
 \end{aligned}$$

Derivada de la función de activación simétrica (2/2)

$$\begin{aligned}\frac{\partial y_j(n)}{\partial v_j(n)} &= 2 \frac{1}{1 + e^{-v_j(n)}} \left(1 - \frac{1}{1 + e^{-v_j(n)}} \right) \\ &= 2 \frac{y_j(n) + 1}{2} \left(1 - \frac{y_j(n) + 1}{2} \right) \\ &= (y_j(n) + 1) \left(1 - \frac{y_j(n) + 1}{2} \right) \\ &= (y_j(n) + 1) \left(\frac{2 - y_j(n) - 1}{2} \right) \\ &= \frac{1}{2} (y_j(n) + 1) (y_j(n) - 1)\end{aligned}$$

Aplicación del gradiente (caso general)

$$\Delta w_{ji}(n) = \mu \delta_j(n) y_i(n)$$

Gradiente de error local instantáneo: $\delta_j = -\frac{\partial \xi(n)}{\partial y_j(n)} \frac{\partial y_j(n)}{\partial v_j(n)}$

$$\delta_j = \frac{\partial \xi(n)}{\partial y_j(n)} \frac{1}{2} (1 + y_j(n))(1 - y_j(n))$$

Retropropagación en la capa III (salida)

$$\Delta w_{ji}^{III}(n) = \mu \delta_j^{III}(n) y_i^{II}(n)$$

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$$\delta_j^{III}(n) = -\frac{\partial \xi(n)}{\partial e_j(n)} \frac{\partial e_j(n)}{\partial y_j^{III}(n)} \frac{1}{2} (1 + y_j^{III}(n)) (1 - y_j^{III}(n))$$

Retropropagación en la capa III (salida)

$$\delta_j^{III}(n) = -\frac{\partial \left\{ \frac{1}{2} \sum_j e_j^2(n) \right\}}{\partial e_j(n)} \cdot \frac{\partial \left\{ d_j^{III}(n) - y_j^{III}(n) \right\}}{\partial y_j^{III}(n)} \cdot \frac{1}{2}(1 + y_j^{III}(n))(1 - y_j^{III}(n))$$

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$$\delta_j^{III}(n) = \frac{1}{2}e_j(n)(1 + y_j^{III}(n))(1 - y_j^{III}(n)) \star$$

Retropropagación en la capa III (salida)

$$\delta_j^{III}(n) = -\frac{\partial \left\{ \frac{1}{2} \sum_j e_j^2(n) \right\}}{\partial e_j(n)} \cdot \frac{\partial \left\{ d_j^{III}(n) - y_j^{III}(n) \right\}}{\partial y_j^{III}(n)} \cdot \frac{1}{2}(1 + y_j^{III}(n))(1 - y_j^{III}(n))$$

$$\delta_j^{III}(n) = \frac{1}{2}e_j(n)(1 + y_j^{III}(n))(1 - y_j^{III}(n)) \star$$

$$\Delta w_{ji}^{III}(n) = \eta e_j(n)(1 + y_j^{III}(n))(1 - y_j^{III}(n))y_i^{II}(n)$$

Propagación hacia atrás: capas ocultas

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Retropropagación en la capa II (oculta)

$$\Delta w_{ji}^{II}(n) = \mu \delta_j^{II}(n) y_i^I(n)$$

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Retropropagación en la capa II (oculta)

$$\Delta w_{ji}^{II}(n) = \mu \delta_j^{II}(n) y_i^I(n)$$

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$$\delta_j^{II}(n) = -\frac{\partial \left\{ \frac{1}{2} \sum_k e_k^2(n) \right\}}{\partial y_j^{II}(n)} \frac{1}{2} (1 + y_j^{II}(n))(1 - y_j^{II}(n))$$

Retropropagación en la capa II (oculta)

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$$\delta_j^{II}(n) = -\frac{\partial \left\{ \frac{1}{2} \sum_k e_k^2(n) \right\}}{\partial y_j^{II}(n)} \frac{1}{2} (1 + y_j^{II}(n))(1 - y_j^{II}(n))$$

$$\delta_j^{II}(n) = -\frac{1}{2} \sum_k \frac{\partial e_k^2(n)}{\partial y_j^{II}(n)} \frac{1}{2} (1 + y_j^{II}(n))(1 - y_j^{II}(n))$$

$$\delta_j^{II}(n) = -\sum_k e_k(n) \frac{\partial e_k(n)}{\partial y_j^{II}(n)} \frac{1}{2} (1 + y_j^{II}(n))(1 - y_j^{II}(n))$$

Retropropagación en la capa II (oculta)

$$\delta_j^H(n) = - \sum_k e_k(n) \frac{\partial e_k(n)}{\partial y_k^H(n)} \frac{\partial y_k^H(n)}{\partial v_k^H(n)} \frac{\partial v_k^H(n)}{\partial y_j^H(n)} \frac{1}{2} (1 + y_j^H(n)) (1 - y_j^H(n))$$

Retropropagación en la capa II (oculta)

$$\delta_j^H(n) = - \sum_k e_k(n) \frac{\partial e_k(n)}{\partial y_k^H(n)} \frac{\partial y_k^H(n)}{\partial v_k^H(n)} \frac{\partial v_k^H(n)}{\partial y_j^H(n)} \frac{1}{2} (1 + y_j^H(n))(1 - y_j^H(n))$$

$$\delta_j^H(n) = - \sum_k e_k(n) \cdot \frac{\partial \{d_k^H(n) - y_k^H(n)\}}{\partial y_k^H(n)} \cdot \frac{1}{2} (1 + y_k^H(n))(1 - y_k^H(n)) \cdot$$

$$\cdot \frac{\partial \left\{ \sum_j w_{kj}^H y_j^H(n) \right\}}{\partial y_j^H(n)} \cdot \frac{1}{2} (1 + y_j^H(n))(1 - y_j^H(n))$$

Retropropagación en la capa II (oculta)

$$\delta_j^H(n) = - \sum_k e_k(n) \frac{\partial e_k(n)}{\partial y_k^H(n)} \frac{\partial y_k^H(n)}{\partial v_k^H(n)} \frac{\partial v_k^H(n)}{\partial y_j^H(n)} \frac{1}{2} (1 + y_j^H(n)) (1 - y_j^H(n))$$

$$\delta_j^H(n) = - \sum_k e_k(n) \cdot \frac{\partial \{d_k^H(n) - y_k^H(n)\}}{\partial y_k^H(n)} \cdot \frac{1}{2} (1 + y_k^H(n)) (1 - y_k^H(n)) \cdot$$

$$\cdot \frac{\partial \{ \sum_j w_{kj}^H y_j^H(n) \}}{\partial y_j^H(n)} \cdot \frac{1}{2} (1 + y_j^H(n)) (1 - y_j^H(n))$$

$$\delta_j^H(n) = - \sum_k e_k(n) \cdot (-1) \cdot \frac{1}{2} (1 + y_k^H(n)) (1 - y_k^H(n)) \cdot$$

$$\cdot w_{kj}^H \cdot \frac{1}{2} (1 + y_j^H(n)) (1 - y_j^H(n))$$

Retropropagación en la capa II (oculta)

$$\delta_j^I(n) = \sum_k e_k(n) \cdot \frac{1}{2}(1 + y_k^{III}(n))(1 - y_k^{III}(n)) \cdot w_{kj}^{III} \cdot \frac{1}{2}(1 + y_j^I(n))(1 - y_j^I(n))$$

Retropropagación en la capa II (oculta)

$$\delta_j^{II}(n) = \sum_k e_k(n) \cdot \frac{1}{2}(1 + y_k^{III}(n))(1 - y_k^{III}(n)) \cdot w_{kj}^{III} \cdot \frac{1}{2}(1 + y_j^{II}(n))(1 - y_j^{II}(n))$$

Pero de la capa III★ sabemos que:

$$\delta_k^{III}(n) = \frac{1}{2}e_k(n)(1 + y_k^{III}(n))(1 - y_k^{III}(n))$$

Retropropagación en la capa II (oculta)

$$\delta_j^H(n) = \sum_k e_k(n) \cdot \frac{1}{2}(1 + y_k^{III}(n))(1 - y_k^{III}(n)) \cdot w_{kj}^{III} \cdot \frac{1}{2}(1 + y_j^H(n))(1 - y_j^H(n))$$

Pero de la capa III★ sabemos que:

$$\delta_k^{III}(n) = \frac{1}{2}e_k(n)(1 + y_k^{III}(n))(1 - y_k^{III}(n))$$

Reemplazando:

$$\delta_j^H(n) = \sum_k \delta_k^{III}(n)w_{kj}^{III} \cdot \frac{1}{2}(1 + y_j^H(n))(1 - y_j^H(n))$$

Retropropagación en la capa II (oculta)

Volviendo a:

$$\Delta w_{ji}^H(n) = \mu \delta_j^H(n) y_i^I(n)$$

Retropropagación en la capa II (oculta)

Volviendo a:

$$\Delta w_{ji}^I(n) = \mu \delta_j^I(n) y_i^I(n)$$

Por lo tanto:

$$\Delta w_{ji}^I(n) = \eta \left[\sum_k \delta_k^{III} w_{kj}^{III}(n) \right] (1 + y_j^I(n))(1 - y_j^I(n)) y_i^I(n)$$

Generalizando para la capa “ p ”

$$\Delta w_{ji}^H(n) = \eta \left[\sum_k \delta_k^H w_{kj}^H(n) \right] (1 + y_j^H(n))(1 - y_j^H(n)) y_i^I(n)$$

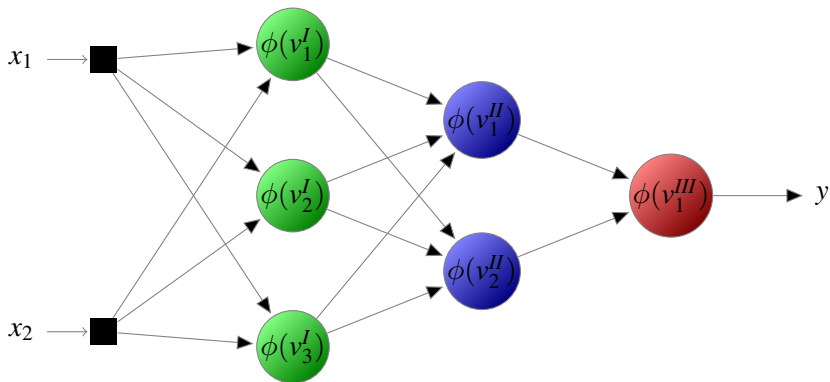
↓

$$\Delta w_{ji}^{(p)}(n) = \eta \left\langle \delta^{(p+1)}, \mathbf{w}_j^{(p+1)} \right\rangle (1 + y_j^{(p)}(n))(1 - y_j^{(p)}(n)) y_i^{(p-1)}(n)$$

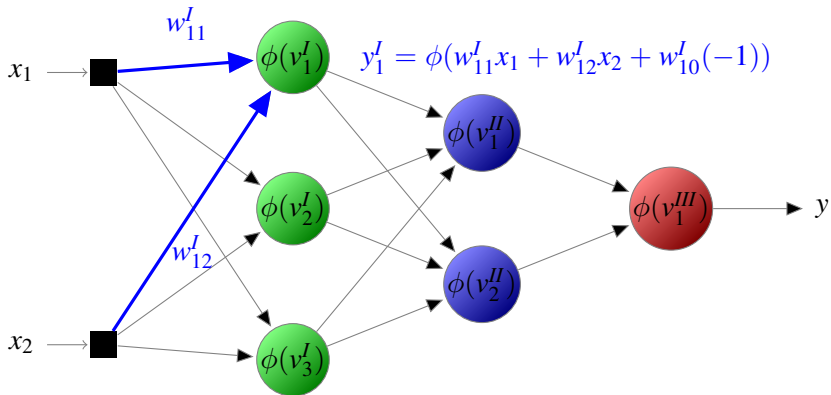
Resumen del algoritmo de retropropagación (BP)

1. Inicialización aleatoria
2. Propagación hacia adelante
3. Propagación hacia atrás
4. Adaptación de los pesos
5. Iteración: vuelve a 2 hasta convergencia o finalización

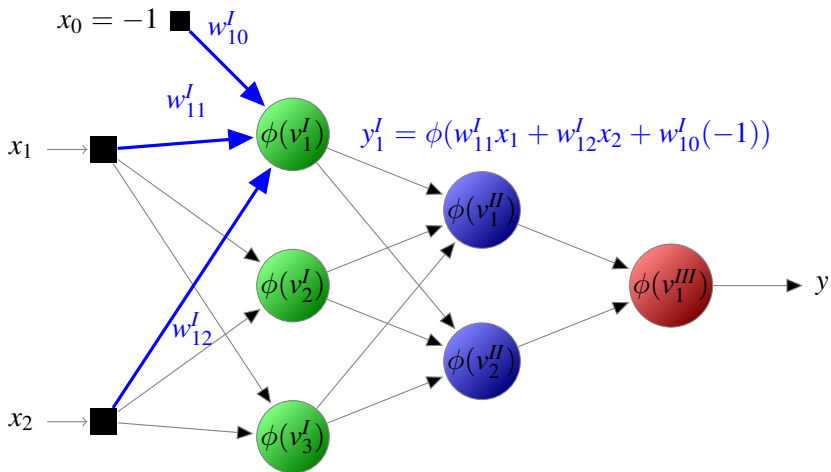
Resumen: propagación hacia adelante



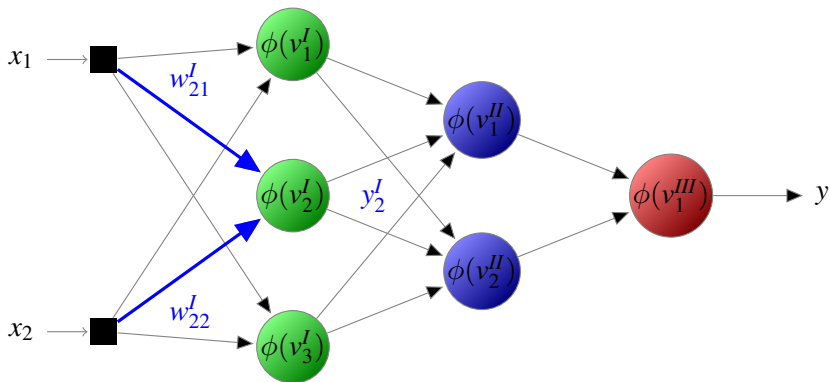
Resumen: propagación hacia adelante



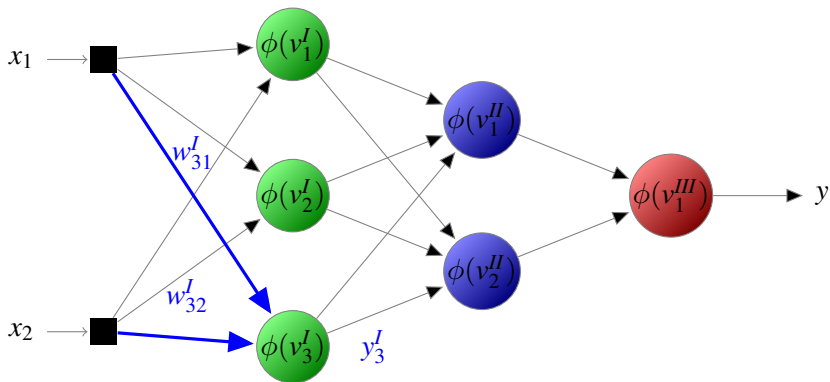
Resumen: propagación hacia adelante



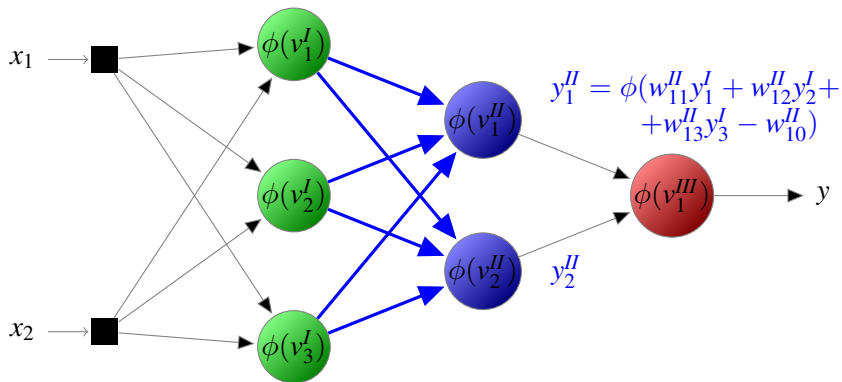
Resumen: propagación hacia adelante



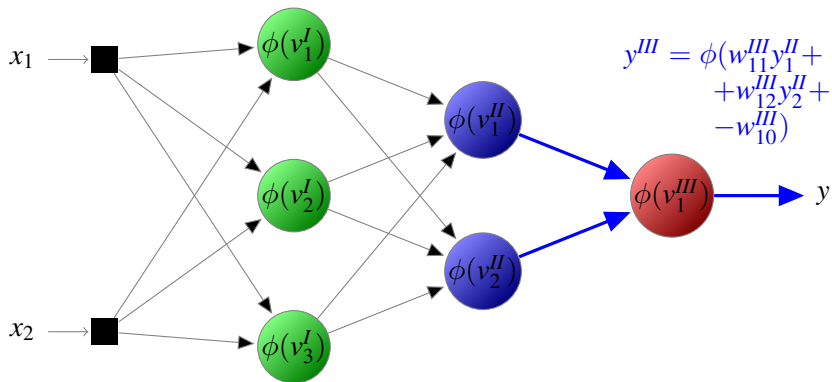
Resumen: propagación hacia adelante



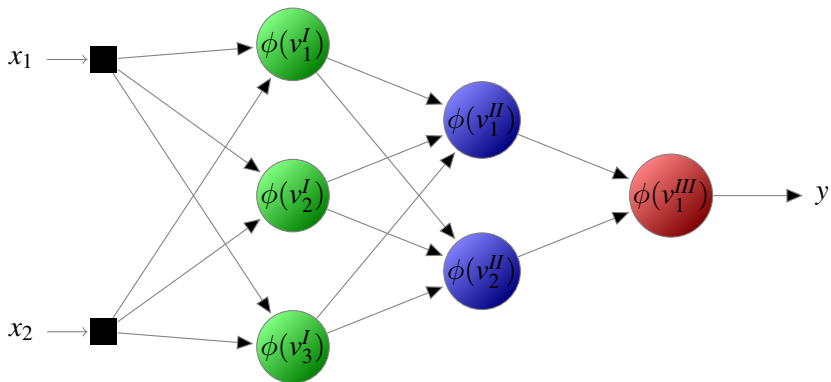
Resumen: propagación hacia adelante



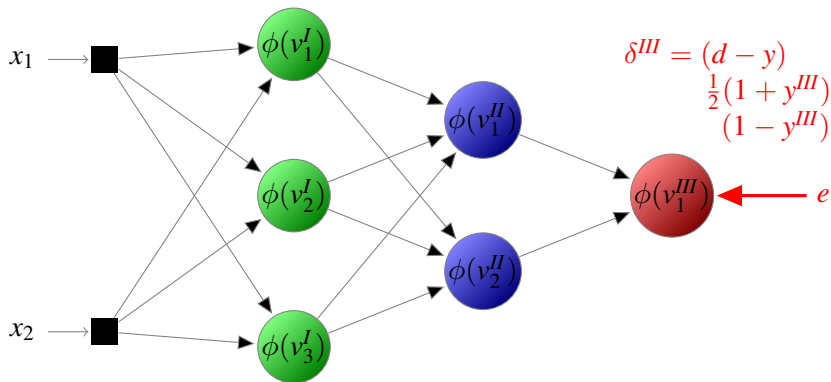
Resumen: propagación hacia adelante



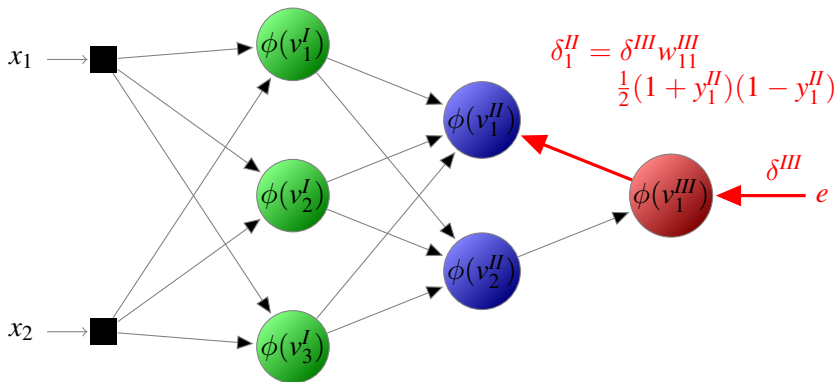
Resumen: propagación hacia **atras**



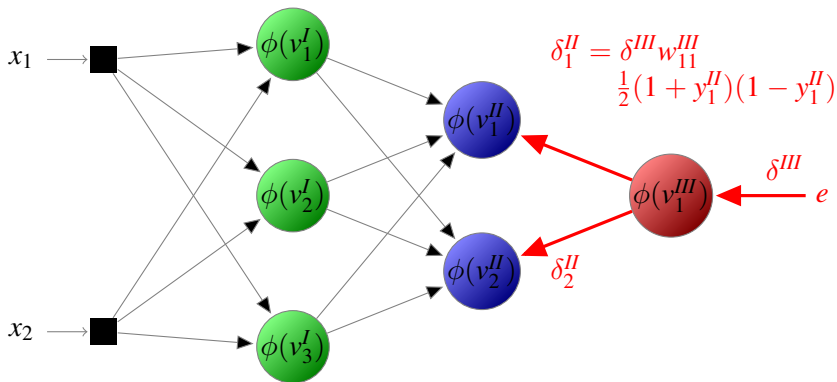
Resumen: propagación hacia **atras**



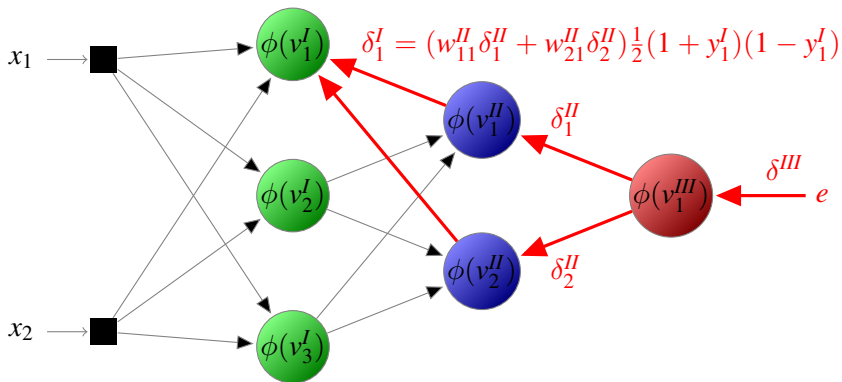
Resumen: propagación hacia **atras**



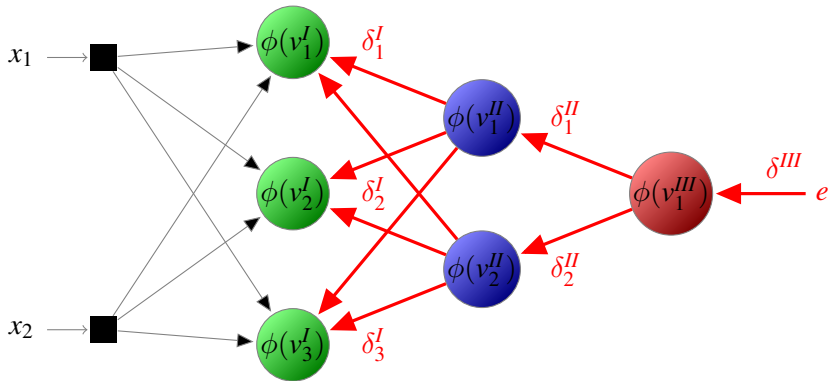
Resumen: propagación hacia **atras**



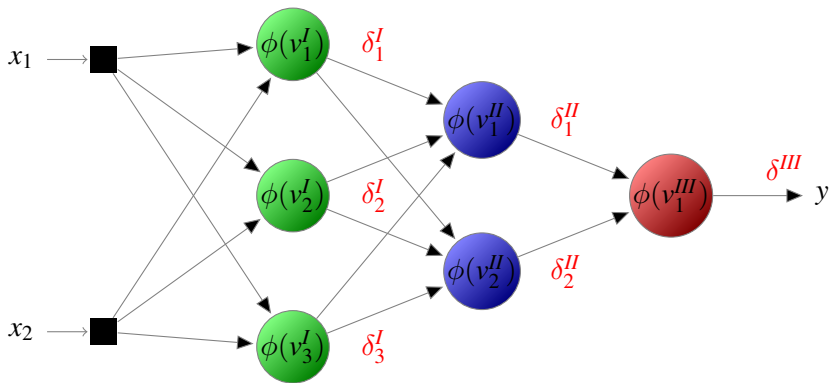
Resumen: propagación hacia **atras**



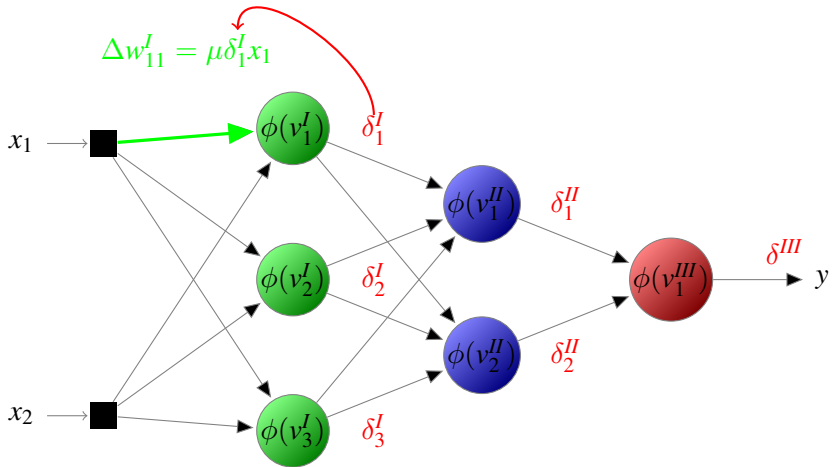
Resumen: propagación hacia **atras**



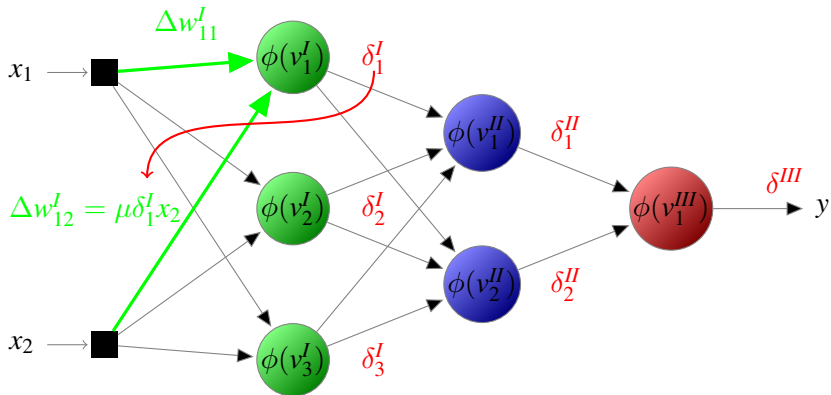
Resumen: ajuste de pesos



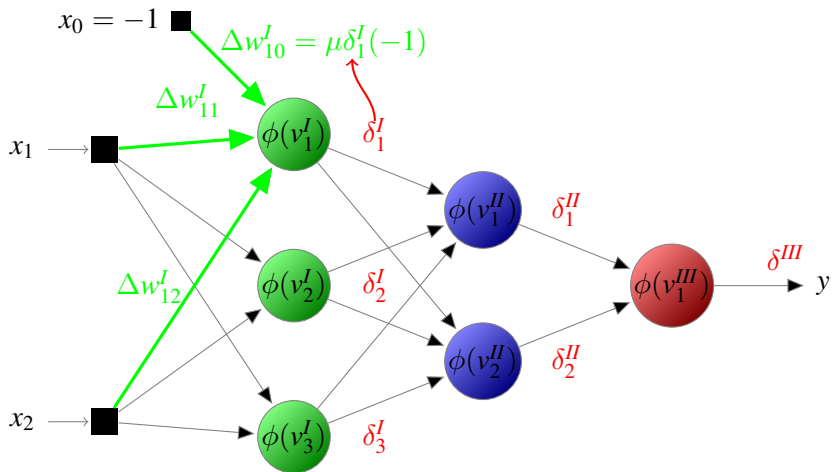
Resumen: ajuste de pesos



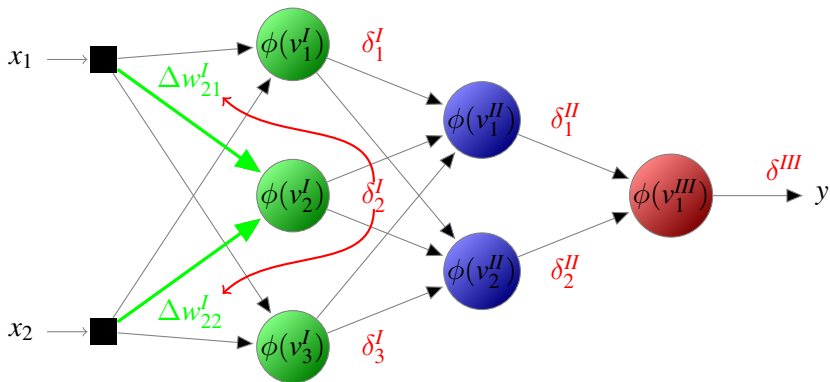
Resumen: ajuste de pesos



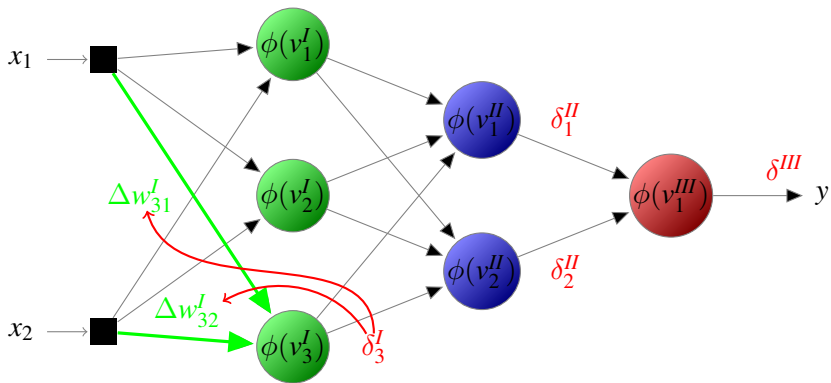
Resumen: ajuste de pesos



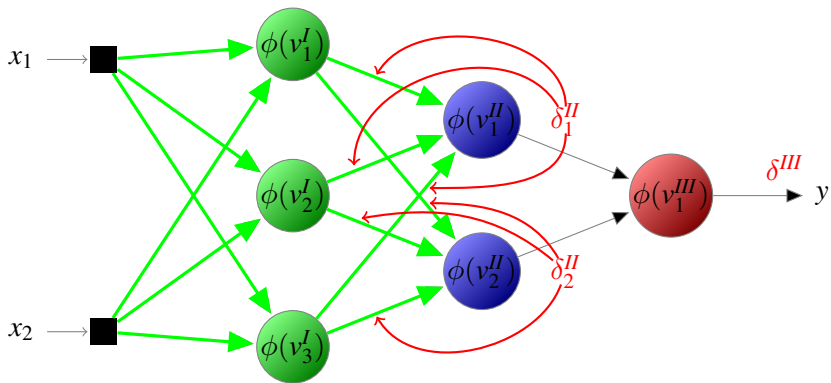
Resumen: ajuste de pesos



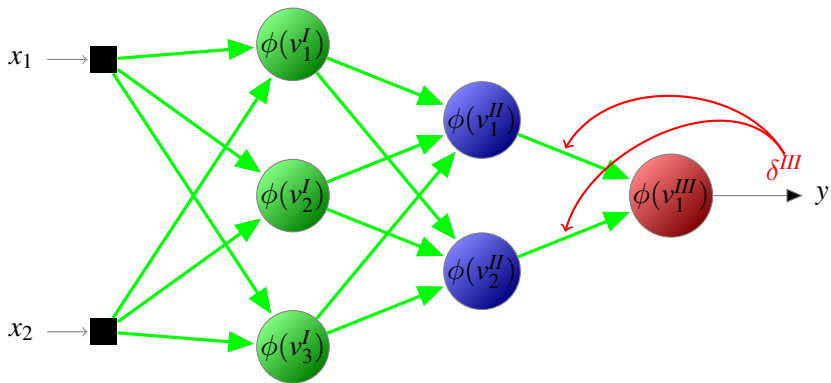
Resumen: ajuste de pesos



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